Pratyus Kumar Panda

me23btech11050@iith.ac.in | pandapratyuskumar@gmail.com | +91 9348297935

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.26
ODM Public School Bhubaneswar	Class XII (CBSE)	2023	95.60%
Modern Public School Balasore	Class X (CBSE)	2021	99.40%

SCHOLASTIC ACHIEVEMENTS

- **NTSE Scholar - 2020/21** : Selected for the National Talent Search Examination scholarship under General Category after clearing both Stage I (State Level) and Stage II (National Level), conducted by NCERT.
- **State Topper - National Standard Examination in Physics 2022/23** : Achieved the highest rank in the state level in the NSEP Olympiad selection stage , conducted by IAPT and HBCSE
- **Reliance Undergraduate Scholar - 2023** : Selected as a Reliance Foundation UG Scholar in the first year for academic excellence; awarded a merit-based scholarship among top-performing students across the country.

EXPERIENCE

Quantitative Strategist Intern

Futures First, Gurgaon

- Analysed high-frequency Australian STIR futures around RBA, CPI, labour-market and GDP events; quantified policy repricing, event-driven volatility and cross-maturity transmission using implied yields, rolling Z-scores, correlation and mean-reversion analysis.
- Constructed fair-value frameworks for SRA/YBA futures from FF/YIB policy strips; decomposed pricing residuals into funding, liquidity, term, credit and basis premia to evaluate rich/cheap and relative-value opportunities.
- Developed and backtested systematic strategies across outrights, calendar spreads and butterflies using statistical diagnostics and regime classification; identified volatility-normalised calendar-spread mean reversion as the most robust framework.
- Applied Bayesian shrinkage to combine historical event studies, RBA guidance and sell-side research into probability-weighted fair values, translating macro scenarios into actionable outright, spread and butterfly trade structures.

SKILLS

- **Programming Languages** : C/C++, Python
- **Product Management & Analytics**: Product Thinking, Customer Segmentation, KPI Tracking, Behavioral Clustering, ETL Pipelines, Dashboarding with Streamlit, Product Roadmapping, MoSCoW Prioritization, User Story Mapping, Product Documentation
- **Mathematical Tools** : Probability, Statistics, Numerical Analysis, Fourier and Laplace Transforms

PROJECTS

Macroeconomic Decision-Support Dashboard

Python, Data Visualisation, Time-Series Analysis, Interactive Dashboarding

- Developed an interactive macro-financial monitoring dashboard integrating inflation, employment, GDP, policy rates, bond yields and equity indices across Australia, US, Japan and New Zealand.
- Designed dynamic filters and time-series views to allow users to quickly compare economic indicators across countries, periods and policy regimes.

- Structured the dashboard as a decision-support interface, reducing the effort required to consolidate macroeconomic information ahead of central-bank meetings and major data releases.
- Enabled rapid identification of economic trends, regime shifts and changing monetary-policy conditions, improving the accessibility of market-relevant macro insights.

Central Bank Communication Intelligence Tool

Python, Text Comparison, NLP-based Classification, Automated Analysis

- Built an automated policy communication analysis tool to compare successive central-bank statements and systematically identify changes in messaging.
- Classified textual changes across key themes including inflation, labour markets and economic growth, creating a structured view of evolving policy priorities.
- Developed a Hawkish/Neutral/Dovish classification framework to translate qualitative central-bank communication into an interpretable measure of policy stance.
- Integrated communication analysis with market expectations to create a repeatable intelligence workflow for assessing how changes in forward guidance could influence decision-making.

Cricket Auction Analytics & Player Valuation Model

Python , EspnCricinfo (Data Source)

- Developed a data-driven auction analytics framework for evaluating cricket players across performance, consistency, role requirements and expected acquisition cost.
- Built a player archetype classification system to segment players into distinct performance profiles, enabling identification of players based on specific squad and tactical requirements rather than overall rankings alone.
- Developed a multi-dimensional performance scoring model using role-specific statistical indicators across batting, bowling and all-round performance to benchmark players against their competitive peer groups.
- Designed a player pricing model that translates performance, consistency and player profile into an estimated auction valuation, supporting bid prioritization and value-for-money decisions.
- Integrated player profiling, valuation and auction strategy into a decision-support framework for identifying target players, setting bid limits and constructing a balanced squad under budget constraints.

POSITIONS OF RESPONSIBILITY

Head of Operations, EBSB, IITH

- **Led end-to-end operational planning and execution**, standardizing workflows through structured checklists, task ownership and volunteer tracking; managed cross-functional teams to deliver high-quality events under tight timelines.
- **Managed event finances and resource allocation**, overseeing budgeting, expense tracking and vendor coordination while optimizing utilization of organizational resources and maintaining operational accountability.

EXTRACURRICULAR

- Active cricket player as an off spin bowling all rounder.
- Published cricket analysis and opinion pieces on Quora, covering player performances, tactics, match analysis and trends in the sport.